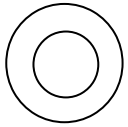
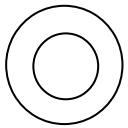
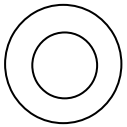
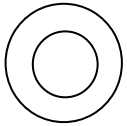
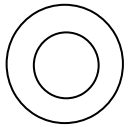
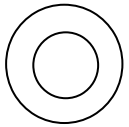
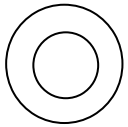
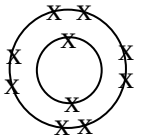
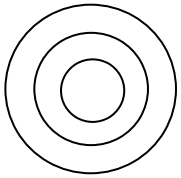
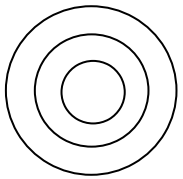
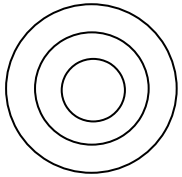
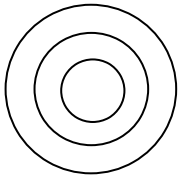
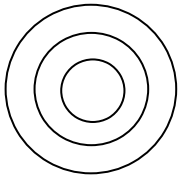
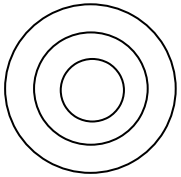
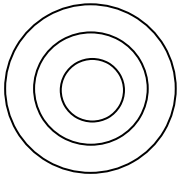
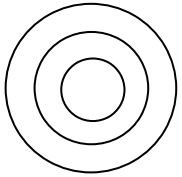
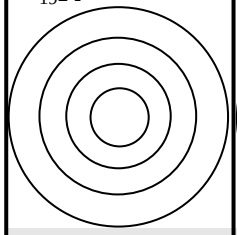
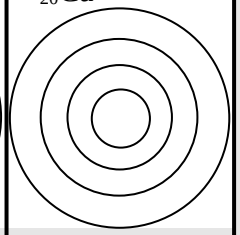
Electronic structure of the first twenty elements in the Periodic Table

Group I	Group II		Group III	Group IV	Group V	Group VI	Group VII	Group 0
$^{1}_{1}H$ 								$^{4}_{2}He$ 
$^{7}_{3}Li$ 	$^{9}_{4}Be$ 		$^{11}_{5}B$ 	$^{12}_{6}C$ 	$^{14}_{7}N$ 	$^{16}_{8}O$ 	$^{19}_{9}F$ 	$^{20}_{10}Ne$  **2,8**
$^{23}_{11}Na$ 	$^{24}_{12}Mg$ 		$^{27}_{13}Al$ 	$^{28}_{14}Si$ 	$^{31}_{15}P$ 	$^{32}_{16}S$ 	$^{35}_{17}Cl$ 	$^{40}_{18}Ar$ 
$^{39}_{19}K$ 	$^{40}_{20}Ca$ 	TRANSITION METALS	Ga	Ge	As	Se	Br	Kr
Rb	Sr		In	Sn	Sb	Te	I	Xe

1. Draw the electronic structure for each element (this is shown for neon)
2. In the grey area under each structure write out the electronic structure (this is shown for neon – **2,8**)

Questions – What do the elements in each Group have in common?

What do the elements in each Period (row) have in common?

Draw and write out the electronic structure for a) a sodium ion b) a chloride ion